Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 6

Title: HR Analytics Dashboard

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HR Analytics Dashboard

Problem Statement

You are an HR Data Analyst in a multinational organization. The HR department wants insights into employee demographics, hiring trends, attrition, and workforce distribution. Develop an interactive HR Analytics Dashboard to support workforce planning and employee management.

Methodology / Steps Followed

Step 1: Data Collection

- Obtained the **IBM HR Analytics Employee Attrition & Performance Dataset** in CSV format.
- Verified that the dataset contained employee information such as age, gender, department, job role, education, monthly income, business travel, marital status, years at company, and attrition.
- The dataset was obtained from the Kaggle IBM HR Analytics dataset.

Step 2: Data Import

- Opened Power BI Desktop.
- Imported the IBM HR Analytics CSV dataset using the **Get Data → Text/CSV** option.
- Loaded the dataset into Power BI for analysis.

Step 3: Data Cleaning and Transformation

- Checked the dataset for missing values and duplicate records.
- Verified and corrected the data types of numerical, categorical, and date-related fields.
- Checked employee attributes such as Age, Monthly Income, Total Working Years, Years at Company, Department, Job Role, Gender, and Attrition.
- Created an additional Age Group column to categorize employees into different age ranges.
- Categorized employees into the following age groups:
- 18–25
- 26–35
- 36–45
- 46–55
- 56+

Step 4: Data Modeling

- Verified data integrity and field relationships within the dataset.
- Created DAX measures for important HR KPIs including:
    - Total Employees
    - Attrition Count
    - Active Employees

- Attrition Rate
- Average Age
- Average Monthly Income

DAX Measures :

- **Total Employees**
  Total Employees = COUNTROWS('project_6_DAB')

- **Attrition Count**
  Attrition Count
  =CALCULATE(COUNTROWS('project_6_DAB'),'project_6_DAB'[Attrition]
  "Yes")

- **Active Employees**
  Active Employees = [Total Employees] - [Attrition Count]

- **Attrition Rate**
  Attrition Rate % = DIVIDE([Attrition Count],[Total Employees],0)

- **Average Age**
  Average Age = AVERAGE('project_6_DAB'[Age])

- **Average Monthly Income**
  Average Monthly Income = AVERAGE('project_6_DAB'[MonthlyIncome])

relationships | measure measure column table | table

Structure | Relationships | Calculations | Calendars

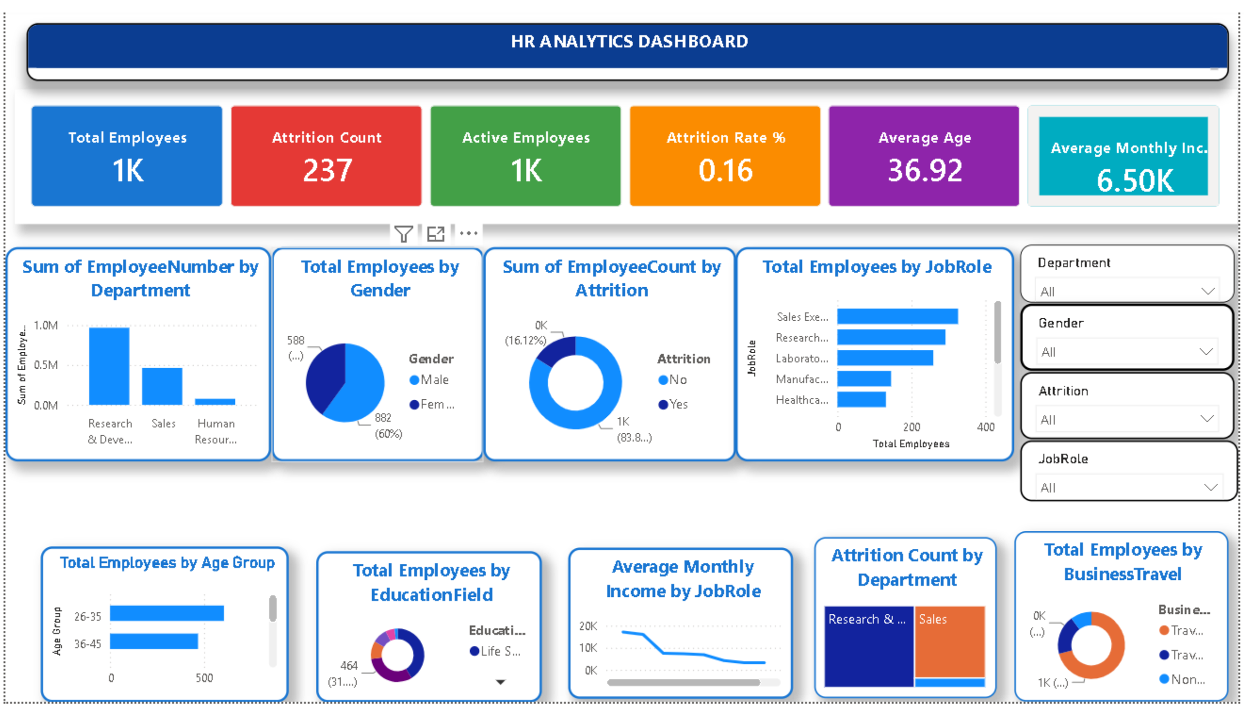
Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber	EnvironmentSatisfaction	Gender	HourlyRate	JobInv
22	No	Non-Travel	1123	Research & Development	16	2	Medical	1	22	4	Male	96	
39	Yes	Travel_Rarely	895	Sales	5	3	Technical Degree	1	42	4	Male	56	
35	No	Travel_Rarely	464	Research & Development	4	2	Other	1	53	3	Male	75	
27	No	Travel_Rarely	1240	Research & Development	2	4	Life Sciences	1	54	4	Female	33	
48	Yes	Travel_Rarely	626	Research & Development	1	2	Life Sciences	1	64	1	Male	98	
45	No	Travel_Rarely	1339	Research & Development	7	3	Life Sciences	1	86	2	Male	59	
36	Yes	Travel_Rarely	318	Research & Development	9	3	Medical	1	90	4	Male	79	
36	No	Travel_Rarely	132	Research & Development	6	3	Life Sciences	1	97	2	Female	55	
32	No	Travel_Rarely	827	Research & Development	1	1	Life Sciences	1	134	4	Male	71	
37	No	Non-Travel	1040	Research & Development	2	2	Life Sciences	1	139	3	Male	100	
34	No	Travel_Rarely	1031	Research & Development	6	4	Life Sciences	1	151	3	Female	45	
19	No	Travel_Rarely	1181	Research & Development	3	1	Medical	1	201	2	Female	79	
51	No	Travel_Rarely	1169	Research & Development	7	4	Medical	1	211	2	Male	34	
38	No	Travel_Rarely	1261	Research & Development	2	4	Life Sciences	1	271	4	Male	88	
38	Yes	Travel_Rarely	1180	Research & Development	29	1	Medical	1	282	2	Male	70	
41	No	Travel_Rarely	896	Sales	6	3	Life Sciences	1	298	4	Female	75	
59	No	Travel_Rarely	142	Research & Development	3	3	Life Sciences	1	309	3	Male	70	
41	No	Travel_Rarely	1411	Research & Development	19	2	Life Sciences	1	334	3	Male	36	
37	Yes	Travel_Frequently	504	Research & Development	10	3	Medical	1	342	1	Male	61	
51	No	Travel_Rarely	833	Research & Development	1	3	Life Sciences	1	353	3	Male	96	
36	No	Travel_Frequently	566	Research & Development	18	4	Life Sciences	1	407	3	Male	81	
39	Yes	Travel_Rarely	1162	Sales	3	2	Medical	1	445	4	Female	41	
43	No	Travel_Rarely	1001	Research & Development	7	3	Life Sciences	1	451	3	Female	43	
42	No	Travel_Rarely	810	Research & Development	23	5	Life Sciences	1	468	1	Female	44	
36	No	Non-Travel	845	Sales	1	5	Medical	1	479	4	Female	45	
33	Yes	Travel_Rarely	350	Sales	5	3	Marketing	1	485	4	Female	34	
46	No	Non-Travel	1144	Research & Development	7	4	Medical	1	487	3	Female	30	

Step 5: Dashboard Development

- Created KPI cards for:
  - Total Employees
  - Attrition Count
  - Active Employees
  - Attrition Rate
  - Average Age
  - Average Monthly Income
- Added slicers for:
  - Department
  - Gender
  - Job Role
  - Attrition Rate
- Developed interactive visualizations including:
  - Employees by Department – Clustered Column Chart
  - Gender Distribution – Pie Chart
  - **Attrition Status** – Donut Chart
  - **Employees by Job Role** – Bar Chart
  - **Employees by Age Group** – Column Chart
  - **Education Field Distribution** – Donut Chart
  - **Average Income by Job Level** – Line Chart
  - **Department-wise Attrition** – Treemap
  - **Business Travel Distribution** – Donut Char

- Applied professional formatting, color themes, chart titles, data labels, borders, and interactive filters to create a user-friendly HR dashboard.

Output



An interactive HR Analytics Dashboard was successfully developed in Power BI using the IBM HR Analytics Employee Attrition & Performance dataset.

The dashboard provides an overview of employee demographics, workforce distribution, attrition, job roles, education, income, and business travel patterns through KPI cards and multiple interactive visualizations.

Users can analyze employee distribution across departments and job roles, compare male and female employees, monitor attrition, examine different age groups, analyze education fields, and compare income across job levels.

Interactive slicers for Department, Gender, Attrition, Job Role, Education Field, Business Travel, Marital Status, and Age Group enable dynamic filtering, making the dashboard user-friendly and effective for HR analysis.

Conclusion

The HR Analytics Dashboard was successfully designed and implemented in Power BI using the IBM HR Analytics dataset.

The dashboard combines KPI cards, interactive charts, and slicers to present meaningful insights from employee data. It enables HR teams to monitor workforce distribution, analyze employee attrition, compare departments and job roles, understand employee demographics, and evaluate income-related patterns.

Overall, the dashboard transforms raw employee data into a clear, interactive, and decision-oriented HR report. It can support HR managers in workforce planning, employee retention, recruitment analysis, and employee management.

Learning Outcomes

After completing this experiment, I learned:

- After completing this experiment, I learned:
- How to import and prepare HR data in Power BI.
- How to clean and transform employee data for analysis.
- How to create calculated columns using DAX.
- How to create DAX measures for HR KPI calculations.
- How to develop interactive KPI cards and HR dashboards.
- How to create Column, Bar, Pie, Donut, Line, and Treemap charts.
- How to use slicers for interactive dashboard filtering.
- How to apply professional formatting and dashboard design principles.
- How to analyze employee demographics and workforce distribution.
- How to analyze employee attrition using Power BI.
- How to identify department-wise and job-role-wise employee patterns.
- How to present HR insights that support data-driven workforce planning and employee management.

GIT HUB LINK :
https://github.com/Balajivallepu/Data_Analytics_for_Bussiness/blob/main/EXPERIMENT 6/Project6updated.pbix